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Electrostatics · Lecture 1

Electric Charge & Coulomb's Law

class notes · Adithya Vikram

1. Electric charge

- Charge = basic property of matter (like mass). It's what causes electric force.
- Only 2 types: positive (+) and negative (−). Names given by Franklin, just labels.
- SI unit → **coulomb (C)**. Small units: 1 $\mu C = 10^{-6}$ C, 1 $nC = 10^{-9}$ C.



like → repel unlike → attract

the golden rule ↑

Where does charge come from?

- atom = nucleus (protons +, neutrons 0) + electrons (−)
- $e = 1.6 \times 10^{-19}$ C (charge of electron is −e, proton is +e)
- $m_e = 9.1 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg
- gains e^- → −ve loses e^- → +ve (only electrons move!)

> **Careful!** +ve body has **LOST electrons**. It did not gain protons.

2. Properties of charge

(a) Additive → it's a scalar

Total charge = algebraic sum. e.g. +3 C, −5 C, +1 C → total = **−1 C**

(b) Quantised

$q = \pm n e \qquad (n = 1, 2, 3, \ldots)$

you can never have 'half an electron' of charge

Q1. A body has charge −3.2 μC. How many extra electrons?

Soln:

1. $n = q / e$
2. $n = \dfrac{3.2 \times 10^{-6}}{1.6 \times 10^{-19}}$
3. $= 2 \times 10^{-6 + 19} = 2 \times 10^{13}$

$\Rightarrow$ **$n = 2 \times 10^{13}$ electrons**

(c) Conserved

Charge can't be created or destroyed, only transferred. Glass + silk: glass loses exactly as many e^- as silk gains.

(d) Invariant

Charge doesn't change with speed (mass does at very high speed).

Charge	Mass
+, − or 0	always +
quantised (n e)	not like that
same at any speed	changes near speed of light
like repel, unlike attract	always attract

3. Conductors vs insulators

- **Conductors**: free electrons → charge moves easily. Metals, human body, earth.
- **Insulators**: electrons tightly bound → charge stays where you put it. Glass, plastic, rubber, silk.
- **Earthing**: connect to Earth. Earth is huge, so it can give/take any charge.

★ **IMP** Charge given to a conductor spreads on its **outer surface**.

4. Ways of charging

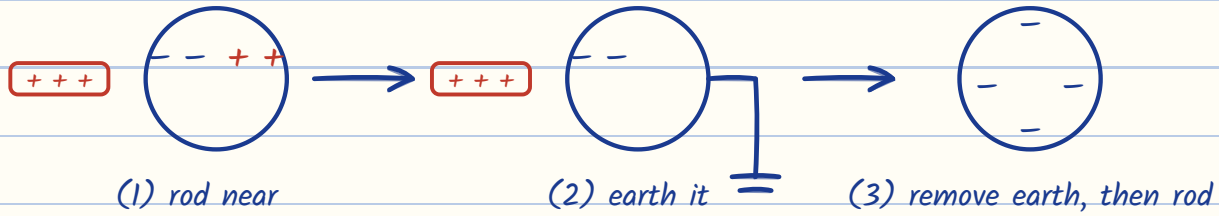
(i) Friction

- glass + silk → glass **+**, silk **−**
- ebonite + fur → ebonite **−**, fur **+**
- both get **equal & opposite** charge

(ii) Conduction (touching)

charged body touches neutral one → neutral one gets **same sign** charge.

(iii) Induction (no touching!)



(1) rod near (2) earth it (3) remove earth, then rod

⇒ sphere is −ve

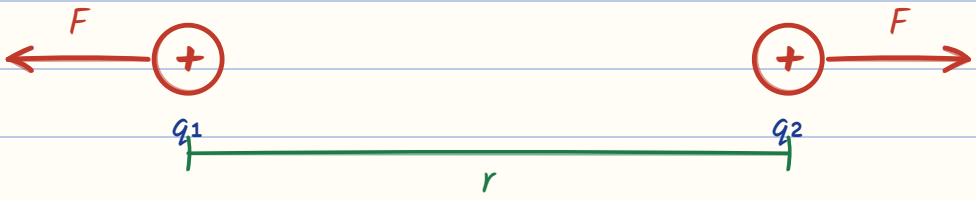
★ **IMP** Induction gives **OPPOSITE** sign, and the rod loses nothing.

Careful! Remove earth FIRST, then the rod. Otherwise the e^- go back to earth → neutral.

5. Coulomb's law (1785)

Force between 2 point charges:

- ∝ product of charges ($q_1 q_2$)
- ∝ $1 / r^2$ (inverse square)
- acts along the line joining them



$$F = k \frac{q_1 q_2}{r^2} \qquad k = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$$

$\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2/N\ m^2$ (permittivity of free space)

- double one charge → F doubles
- double r → F becomes ¼ (NOT ½!)
- put signs in: answer + → repulsion, − → attraction
- valid for point charges at rest (electro-statics)

Vector form

$$\vec{F}_{21} = k \frac{q_1 q_2}{r^3} \vec{r}_{12}$$

force on 2 due to 1; $\vec{r}_{12}$ points from 1 to 2. And $\vec{F}_{12} = -\vec{F}_{21}$

Q2. Force between +3 μC and −2 μC, 30 cm apart (vacuum)?

Sol^n:

1. $q_1 = 3\times10^{-6}$, $q_2 = 2\times10^{-6}$, $r = 0.3$ m

2. $F = \dfrac{9\times10^9 \times 3\times10^{-6} \times 2\times10^{-6}}{(0.3)^2}$

3. $= \dfrac{54 \times 10^{-3}}{0.09} = 0.6$ N

4. opposite signs → attraction

⇒ **F = 0.600 N (attractive)**

6. In a medium

$$F_{medium} = \frac{F_{vacuum}}{K} \qquad \varepsilon = K\,\varepsilon_0$$

K = dielectric constant (relative permittivity)

- vacuum $K = 1$, air ≈ 1, water ≈ 80, metal $\to \infty$

Same charges as above, now in a liquid with $K = 4 \to F = 0.6 / 4 =$ **0.150 N**

7. Electric vs gravity

Electric	Gravitational
$F = k\,q_1q_2/r^2$	$F = G\,m_1m_2/r^2$
attract or repel	only attract
depends on medium	doesn't
very strong	very weak

Q3. Ratio F_e/F_g for an electron and a proton?

Soln:

1. r^2 cancels! ratio $= \dfrac{k\,e^2}{G\,m_e\,m_p}$

2. $= \dfrac{(9\times10^{9})(1.6\times10^{-19})^2}{(6.67\times10^{-11})(9.1\times10^{-31})(1.67\times10^{-27})}$

$\Rightarrow$ **$\approx 2.27 \times 10^{39}$** $\to$ **electric force is HUGE**

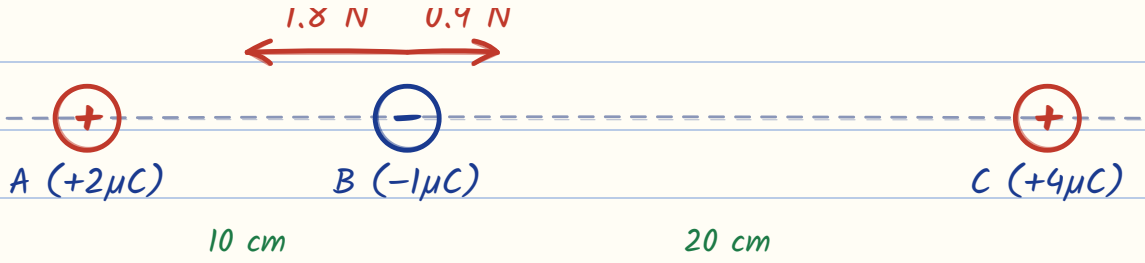
Trick Gravity rules space because big bodies are almost neutral; their electric forces cancel.

8. Superposition

Each pair acts **independently**. Net force = **vector sum**:

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

Method: (1) size of each force by Coulomb (2) direction: away if like, towards if unlike (3) add as vectors



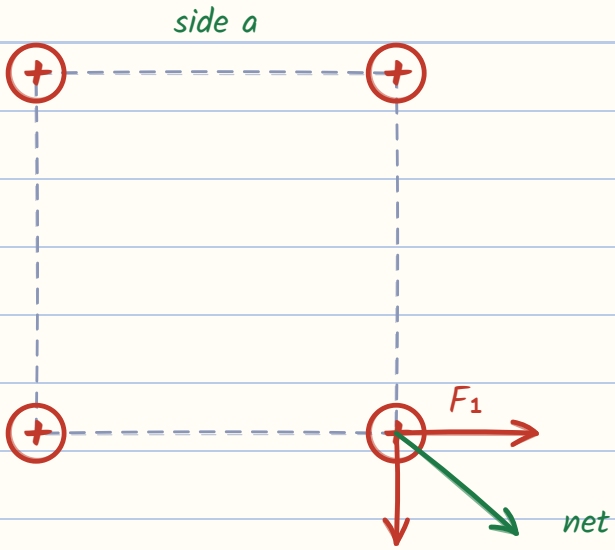
Q4. A = +2 μC (x = 0), B = −1 μC (x = 10 cm), C = +4 μC (x = 30 cm). Net force on B?

Soln:

1. from A: $\frac{9\times10^{9}\times2\times10^{-6}\times1\times10^{-6}}{(0.1)^2}$ = 1.8 N → towards A
2. from C: $\frac{9\times10^{9}\times4\times10^{-6}\times1\times10^{-6}}{(0.2)^2}$ = 0.9 N → towards C
3. opposite directions → 1.8 − 0.9

⇒ **0.900 N towards A**

Four charges on a square



QS. 4 equal charges $q = 1\ \mu C$ on a square of side 10 cm. Net force on one?

Soln:

1. 2 neighbours: each $F_1 = kq^2/a^2 = 0.900$ N, at 90° → $\sqrt{2}\ F_1$
2. diagonal one: distance $\sqrt{2}\ a$ → $F_1/2$, along the same diagonal
3. total = $(\sqrt{2} + ½)\ F_1$

⇒ **$F \approx 1.72$ N along the diagonal**

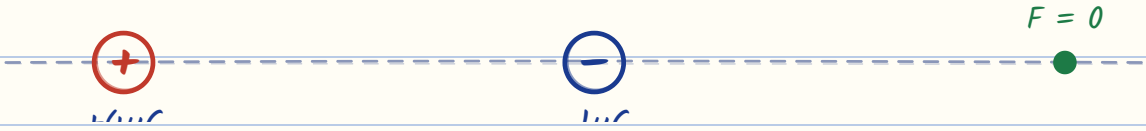
Trick Equilateral triangle (side a, equal q): $F = \sqrt{3} \cdot kq^2/a^2$ on each.

9. Null point (net force = 0)

like charges → null pt. in between, near smaller



unlike charges → null pt. outside, beyond smaller



★ **IMP** like → between, near smaller | unlike → outside, beyond smaller | test charge cancels out

Q6. +2 μC and +8 μC, 30 cm apart. Null point?

Soln:

1. let x from 2 μC → $\frac{2}{x^2} = \frac{8}{(30-x)^2}$
2. √ both sides: $\frac{1}{x} = \frac{2}{30-x}$ → $30 - x = 2x$
3. $x = 10$

⇒ **10.0 cm from +2 μC**

Q7. +4 μC and −1 μC, 30 cm apart. Null point?

Soln:

1. unlike → outside, beyond −1 μC. Let x beyond it.
2. $\frac{4}{(30+x)^2} = \frac{1}{x^2}$ → $2x = 30 + x$

⇒ **x = 30.0 cm beyond −1 μC** (60 cm from +4 μC)

All 3 in equilibrium

q and 4q, distance d. Third charge Q = **−4q/9** at **d/3** from q.

(why? Q sits at the null point → d/3. Then for q: pull of Q = push of 4q → $\frac{kq|Q|}{(d/3)^2} = \frac{kq \cdot 4q}{d^2}$ → $|Q| = 4q/9$, and Q must be −ve)

10. Touching identical spheres

each gets $\frac{q_1 + q_2}{2}$ (with signs!)

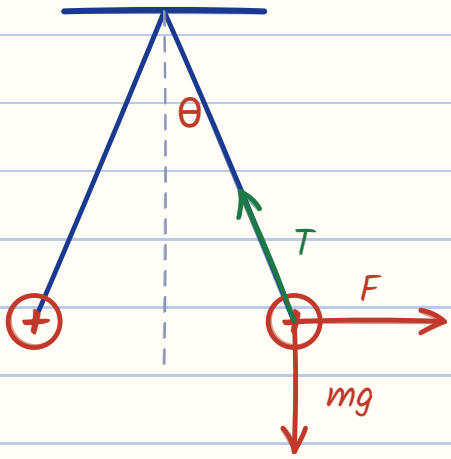
Q8. +8 μC and −2 μC identical spheres touch, then back to same distance. F'/F ?

Soln:

1. each = (8 − 2)/2 = +3 μC
2. before ∝ 8 × 2 = 16 (attract)
3. after ∝ 3 × 3 = 9 (repel)

⇒ **F'/F = 9/16, attraction → repulsion**

11. Hanging charged balls



1. vertical: $T \cos\theta = mg$
2. horizontal: $T \sin\theta = F$
3. divide → T cancels

$$\tan\theta = \frac{F}{mg} = \frac{kq^2}{r^2 mg}$$

e.g. $m = 0.9\ g$, $q = 0.1\ \mu C$, $r = 10\ cm \rightarrow F = 9\times10^{-3}\ N = mg \rightarrow \tan\theta = 1$

$\rightarrow \theta = 45.0°$

★ **IMP** In a liquid (density σ, dielectric K): same angle if $K = \frac{\rho}{\rho - \sigma}$ (ρ = ball density). e.g. $\rho = 1600$, $\sigma = 800 \rightarrow K = 2.00$

12. Max force trick

Q split into q and (Q − q): $F \propto q(Q - q) \rightarrow d/dq = 0 \rightarrow$ **$q = Q/2$**

Common mistakes X

- not converting μC → C and cm → m
- forgetting to square r
- adding forces like numbers (they're **vectors**!)
- "bigger charge feels bigger force" → NO, both feel the same (Newton's 3rd)
- null point on wrong side
- sharing charge without signs (+8, −2 → +3 each, not +5)
- in a medium force is **÷ K**, not × K

Quick formula sheet

$q = n\,e$, $e = 1.6\times10^{-19}$ C

$F = k\frac{q_1 q_2}{r^2}$, $k = 9\times10^9$

$\varepsilon_0 = 8.85\times10^{-12}$

$F_{medium} = F / K$

$\vec{F}_{net} = \Sigma \vec{F}_i$

null pt (like): $x = \frac{d\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}}$

touching: $(q_1 + q_2)/2$

hanging: $\tan\theta = F / mg$

square: $(\sqrt{2} + \frac{1}{2})\,kq^2/a^2$

triangle: $\sqrt{3}\,kq^2/a^2$

— end of lecture 1 —